

Chapter 1: Computer Networks and the Internet

Multiple-Choice Questions (140 Questions)

Section 1.1: What Is the Internet? (Questions 1-20)

1. What is the primary focus of the book when discussing computer networks?

- A) Private corporate networks
- B) The public Internet and its protocols
- C) Local area networks only
- D) Satellite communication systems

Answer: B (*The book uses the public Internet as the principal vehicle for discussing computer networks and protocols.*)

2. How can the Internet be described in terms of its basic components?

- A) Only as a service provider for applications
- B) As hardware and software components (nuts and bolts) or as infrastructure for distributed applications
- C) Solely as a collection of servers
- D) As a single global database

Answer: B (*The Internet can be described via its hardware/software components or as networking infrastructure for services to applications.*)

3. What are computing devices connected to the Internet called in Internet jargon?

- A) Servers
- B) Hosts or end systems
- C) Routers
- D) Links

Answer: B (*All computing devices like desktops, smartphones, and IoT devices are called hosts or end systems.*)

4. According to estimates mentioned, how many devices were connected to the Internet by 2022?

- A) 18 billion
- B) 28.5 billion
- C) 10 billion
- D) 50 billion

Answer: B (*Estimates indicate 28.5 billion devices by 2022, up from 18 billion in 2017.*)

5. What connects end systems in a network?

- A) Only packet switches*
- B) A network of communication links and packet switches*
- C) Servers exclusively*
- D) Protocols alone*

Answer: B (*End systems are connected by communication links and packet switches.*)

6. What is a packet in computer networks?

- A) A complete message sent without segmentation*
- B) Segmented data with added headers sent through the network*
- C) A hardware device*
- D) A type of router*

Answer: B (*Packets are packages of segmented data with headers sent to the destination for reassembly.*)

7. What are the two main types of packet switches in the Internet?

- A) Hosts and end systems*
- B) Routers and link-layer switches*
- C) Servers and clients*
- D) Links and protocols*

Answer: B (*Routers and link-layer switches are the prominent types.*)

8. Where are link-layer switches typically used?

- A) In the network core*
- B) In access networks*
- C) Only in end systems*
- D) In global transit*

Answer: B (*Link-layer switches are used in access networks, while routers are in the core.*)

9. What is the sequence of links and switches a packet traverses called?

- A) A protocol*
- B) A route or path*
- C) A header*
- D) A segment*

Answer: B (*It is known as a route or path through the network.*)

10. How do end systems access the Internet?

- A) Directly through global routers*
- B) Through Internet Service Providers (ISPs)*
- C) Via packet switches only*
- D) Using protocols without intermediaries*

Answer: B (*End systems access via ISPs like residential, corporate, or cellular providers.*)

11. What interconnects lower-tier ISPs?

- A) End systems*
- B) National and international upper-tier ISPs*

C) Only local links

D) Application protocols

Answer: B (*Lower-tier ISPs are interconnected through upper-tier ISPs.*)

12. What are the two most important protocols in the Internet?

A) HTTP and SMTP

B) TCP and IP

C) Ethernet and WiFi

D) DNS and FTP

Answer: B (*TCP and IP are key, collectively known as TCP/IP.*)

13. Who develops Internet standards?

A) IEEE

B) IETF (Internet Engineering Task Force)

C) Cisco

D) Google

Answer: B (*The IETF develops standards documented as RFCs.*)

14. What are RFCs?

A) Hardware specifications

B) Requests for comments that define protocols like TCP and IP

C) Network hardware

D) Application services

Answer: B (*RFCs are IETF standards documents for protocols.*)

15. What body specifies standards for network links like Ethernet?

A) IETF

B) IEEE 802 LAN Standards Committee

C) Cisco VNI

D) Fiber Broadband

Answer: B (*The IEEE 802 committee specifies Ethernet and WiFi standards.*)

16. How is the Internet described as an infrastructure?

A) Only for hardware connections

B) As providing services to distributed applications

C) Solely for packet switching

D) As a single protocol stack

Answer: B (*It provides services to applications like email, web, and streaming.*)

17. What are distributed applications?

A) Programs running on a single system

B) Applications involving multiple end systems exchanging data

C) Only server-based programs

D) Hardware components

Answer: B (*They involve multiple end systems, running on end systems not in the core.*)

18. What interface do end systems use to deliver data over the Internet?

- A) Packet switch interface
- B) Socket interface
- C) Router interface
- D) Link-layer interface

Answer: B (*The socket interface specifies rules for sending data to another program.*)

19. In a human protocol analogy, what initiates communication?

- A) Asking for time directly
- B) Offering a greeting like "Hi"
- C) Sending a packet
- D) Closing the connection

Answer: B (*Human protocol starts with a greeting to initiate communication.*)

20. What defines a network protocol?

- A) Hardware only
- B) Format and order of messages, plus actions on transmission/receipt
- C) Physical media
- D) End systems alone

Answer: B (*It defines message format, order, and actions taken.*)

Section 1.2: The Network Edge (Questions 21-40)

21. What are end systems also referred to as?

- A) Routers
- B) Hosts
- C) Links
- D) Switches

Answer: B (*End systems are hosts that run applications.*)

22. How are hosts categorized?

- A) Into links and switches
- B) Into clients and servers
- C) Into protocols and layers
- D) Into physical media

Answer: B (*Clients are desktops/smartphones; servers store/distribute content.*)

23. Where do most servers for search results and email reside today?

- A) In homes
- B) In large data centers
- C) On mobile devices
- D) In access networks

Answer: B (*They reside in large data centers.*)

24. What is the access network?

- A) *The core of the Internet*
- B) *The network connecting an end system to the first router*
- C) *Only wireless connections*
- D) *Global transit links*

Answer: B (*It physically connects to the edge router.*)

25. What are the two most prevalent broadband residential access types?

- A) *Ethernet and WiFi*
- B) *DSL and cable*
- C) *FTTH and 5G*
- D) *Satellite and radio*

Answer: B (*DSL and cable are the main types.*)

26. In DSL, what device is located in the telco's central office?

- A) *Cable modem*
- B) *DSLAM (Digital Subscriber Line Access Multiplexer)*
- C) *Router*
- D) *Switch*

Answer: B (*DSLAM handles data from DSL modems.*)

27. How are data and telephone signals handled in DSL?

- A) *On separate lines*
- B) *Encoded at different frequencies on the same line*
- C) *Only digitally*
- D) *Via satellite*

Answer: B (*They use different frequencies.*)

28. What is the typical range for DSL to work effectively?

- A) *1-2 miles*
- B) *5-10 miles*
- C) *20 miles*
- D) *Unlimited*

Answer: B (*DSL is designed for short distances up to 5-10 miles from the CO.*)

29. What infrastructure does cable Internet use?

- A) *Telephone lines*
- B) *Cable television infrastructure*
- C) *Fiber optics only*
- D) *Wireless spectrum*

Answer: B (*It uses existing cable TV infrastructure.*)

30. What is HFC in cable access?

- A) *High-frequency coax*
- B) *Hybrid fiber coax*
- C) *Home fiber connection*

D) High-speed fiber

Answer: B (*HFC combines fiber and coaxial cable.*)

31. What device serves a similar function to DSLAM in cable networks?

A) DSL modem

B) CMTS (Cable Modem Termination System)

C) Router

D) Switch

Answer: B (*CMTS converts analog signals to digital.*)

32. Why is cable Internet access shared?

A) It uses dedicated lines

B) Every packet travels on shared links to all homes

C) Only upstream is shared

D) It is not shared

Answer: B (*It is a shared broadcast medium.*)

33. What technology provides gigabit speeds directly to homes?

A) DSL

B) Cable

C) FTTH (Fiber to the Home)

D) Ethernet

Answer: C (*FTTH provides high-speed optical fiber to homes.*)

34. What is the simplest FTTH technology?

A) Hybrid fiber

B) Direct fiber

C) Coaxial fiber

D) Wireless fiber

Answer: B (*Direct fiber from CO to home.*)

35. What is the most prevalent access technology in enterprises?

A) DSL

B) Ethernet

C) Cable

D) Satellite

Answer: B (*Ethernet is prevalent in corporate, university, and home networks.*)

36. What standard is WiFi based on?

A) IEEE 802.3

B) IEEE 802.11

C) IEEE 802.1

D) IEEE 802.15

Answer: B (*WiFi is IEEE 802.11 technology.*)

37. What is the range for a wireless LAN user from an access point?

- A) A few meters
- B) A few tens of meters
- C) Kilometers
- D) Unlimited

Answer: B (*Typically within a few tens of meters.*)

38. What generation of wireless provides wide-area access up to tens of kilometers?

- A) WiFi
- B) Ethernet
- C) 3G, 4G, and 5G cellular
- D) FTTH

Answer: C (*Cellular like 4G/5G allows access within tens of kilometers of base stations.*)

39. What are the two categories of physical media?

- A) Wired and wireless
- B) Guided and unguided
- C) Copper and fiber
- D) Terrestrial and satellite

Answer: B (*Guided media guide waves; unguided propagate freely.*)

40. What is the most common guided medium for LANs?

- A) Fiber optics
- B) Coaxial cable
- C) Twisted-pair copper wire
- D) Radio spectrum

Answer: C (*Twisted-pair is least expensive and commonly used.*)

Section 1.3: The Network Core (Questions 41-65)

41. What breaks messages into packets?

- A) Routers
- B) The source end system
- C) Switches
- D) Links

Answer: B (*The sending end system segments data into packets.*)

42. What is store-and-forward transmission?

- A) Transmitting bits immediately
- B) Receiving the entire packet before forwarding
- C) Segmenting packets
- D) Queuing only

Answer: B (*The switch must receive the whole packet first.*)

43. For a path with N links of rate R and packet length L , what is the end-to-end delay without other delays?

- A) L/R
- B) $N(L/R)$
- C) $(N-1)(L/R)$
- D) $2L/R$

Answer: C (*End-to-end delay is $(N-1)(L/R)$ for store-and-forward.*)

44. What causes queuing delays?

- A) Empty buffers
- B) Packets waiting in output buffers due to congestion
- C) Propagation speed
- D) Header addition

Answer: B (*Packets wait if the link is busy.*)

45. What happens when a queue is full?

- A) Faster transmission
- B) Packet loss
- C) Automatic rerouting
- D) No effect

Answer: B (*Arriving packets are dropped, causing loss.*)

46. How does a router determine where to forward a packet?

- A) Using protocols
- B) Forwarding table based on destination address
- C) Randomly
- D) Via physical media

Answer: B (*It uses a forwarding table mapping addresses to links.*)

47. What sets forwarding tables automatically?

- A) End systems
- B) Routing protocols
- C) Applications
- D) Physical layer

Answer: B (*Routing protocols set the tables.*)

48. In circuit switching, what is reserved for a session?

- A) Packets
- B) Resources like buffers and link rates
- C) Messages
- D) Headers

Answer: B (*Resources are reserved along the path.*)

49. What are the two multiplexing methods in circuit switching?

- A) Packet and message
- B) FDM and TDM

- C) Store-and-forward
- D) Queuing and propagation

Answer: B (*Frequency-division and time-division multiplexing.*)

50. Why is packet switching better for sharing capacity?

- A) It reserves resources
- B) It allows better sharing than circuit switching
- C) It has fixed delays
- D) No queuing

Answer: B (*It offers better sharing and is more efficient.*)

51. What trend is seen in telecommunication networks?

- A) Toward circuit switching
- B) Toward packet switching
- C) Away from multiplexing
- D) To proprietary networks

Answer: B (*Networks are migrating to packet switching.*)

52. In Network Structure 1, how are access ISPs interconnected?

- A) Directly to each other
- B) Via a single global transit ISP
- C) Through end systems
- D) Without interconnection

Answer: B (*All connect to one global transit ISP.*)

53. What is a customer-provider relationship in ISPs?

- A) Free peering
- B) Access ISP pays the global ISP
- C) Equal sharing
- D) No payment

Answer: B (*The access ISP is a customer paying the provider.*)

54. What does Network Structure 2 add?

- A) Single ISP
- B) Multiple competing global transit ISPs
- C) Only access ISPs
- D) End systems

Answer: B (*Multiple global ISPs compete.*)

55. What are tier-1 ISPs?

- A) Local access providers
- B) Global ISPs with presence not in every city
- C) Only regional
- D) Home networks

Answer: B (*Tier-1 like AT&T connect regionally.*)

56. In Network Structure 3, what connects access ISPs in a region?

- A) Global ISPs directly*
- B) Regional ISPs that connect to tier-1*
- C) Peering points*
- D) Content providers*

Answer: B (*Regional ISPs connect access to tier-1.*)

57. What is a PoP?

- A) Point of presence for customer connections*
- B) Protocol over protocol*
- C) Packet of packets*
- D) Point of peering*

Answer: A (*Group of routers for ISP connections.*)

58. What is multi-homing?

- A) Connecting to one ISP*
- B) Connecting to two or more provider ISPs*
- C) Single link use*
- D) No redundancy*

Answer: B (*For redundancy against failures.*)

59. What is peering between ISPs?

- A) Payment-based connection*
- B) Direct, settlement-free connection*
- C) Through tier-1 only*
- D) Via end systems*

Answer: B (*Nearby ISPs connect directly without payment.*)

60. What is an IXP?

- A) Internet exchange point for peering*
- B) ISP expansion protocol*
- C) Internal exchange protocol*
- D) Internet xylem point*

Answer: A (*Meeting point for multiple ISPs to peer.*)

61. What does Network Structure 5 add?

- A) Only access ISPs*
- B) Content-provider networks like Google*
- C) More regional ISPs*
- D) End systems*

Answer: B (*Adds networks like Google's private TCP/IP.*)

62. How does Google bypass upper tiers?

- A) Using public Internet only*
- B) Peering with lower-tier ISPs and IXPs*
- C) Paying all tier-1*

D) No bypassing

Answer: B (*Direct peering to reduce costs.*)

63. What is twisted-pair copper wire used for?

A) Long-haul only

B) LANs and residential access

C) Satellite links

D) Optical pulses

Answer: B (*Common for LANs and DSL.*)

64. What achieves data rates up to 10 Gbps over 100 meters?

A) Coaxial cable

B) Category 6a twisted-pair

C) Radio channels

D) Satellite

Answer: B (*Modern twisted-pair like cat 6a.*)

65. What is fiber optics preferred for?

A) Short-haul LANs

B) Long-haul transmission

C) Wireless

D) Low bit rates

Answer: B (*For overseas and backbone due to high rates.*)

Section 1.4: Delay, Loss, and Throughput (Questions 66-90)

66. What are the four types of nodal delays?

A) Processing, queuing, transmission, propagation

B) Packet, message, segment, frame

C) Link, route, path, switch

D) Header, payload, buffer, bit

Answer: A (*These accumulate to total nodal delay.*)

67. What is included in processing delay?

A) Transmitting bits

B) Examining header and determining outbound link

C) Waiting in queue

D) Propagating over link

Answer: B (*Includes header check and error detection.*)

68. What causes variable queuing delays?

A) Fixed buffer size

B) Number of packets waiting for transmission

C) Constant arrival rate

D) No congestion

Answer: B (Depends on earlier packets and congestion.)

69. What is transmission delay?

A) Time to propagate

B) L/R , time to push all bits into the link

C) Queue wait

D) Header processing

Answer: B (L bits at rate R .)

70. What is propagation delay?

A) d/s , distance over propagation speed

B) Time in queue

C) Bit pushing time

D) Error checking

Answer: A (d distance, s speed near light.)

71. What is the formula for total nodal delay?

A) $d_{proc} + d_{queue} + d_{trans} + d_{prop}$

B) L/R only

C) $N-1$

D) Traffic intensity

Answer: A (Sum of all components.)

72. What is traffic intensity?

A) $\lambda a/R$, arrival rate times packet length over transmission rate

B) End-to-end delay

C) Packet loss probability

D) Throughput

Answer: A ($\lambda a/R$ estimates queuing delay.)

73. What happens if traffic intensity > 1 ?

A) Zero delay

B) Queue grows infinitely

C) No loss

D) Fixed delay

Answer: B (Arrival exceeds transmission.)

74. As traffic intensity approaches 1, what happens to average queuing delay?

A) Decreases

B) Increases rapidly

C) Stays constant

D) Becomes zero

Answer: B (Delay increases sharply.)

75. What causes packet loss in real queues?

- A) Infinite capacity
- B) Finite buffer space leading to drops
- C) No queuing
- D) Automatic retransmission

Answer: B (Full buffer drops packets.)

76. For N-1 routers, uncongested end-to-end delay is?

- A) $N(d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$
- B) $(N-1)(d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$
- C) L/R
- D) d_{queue} only

Answer: A (Processing/transmission at N nodes, propagation on N-1 links, but approximated as such.)

77. What does Traceroute measure?

- A) Throughput
- B) Delays to routers along the path
- C) Packet loss only
- D) Bandwidth

Answer: B (Sends packets to measure round-trip delays.)

78. What can add delays in end systems?

- A) Only nodal delays
- B) Application processing like media encoding
- C) No delays
- D) Physical media

Answer: B (e.g., Encoding/decoding in VoIP.)

79. What is throughput?

- A) Delay measure
- B) Rate of data transfer from source to destination
- C) Packet loss
- D) Queuing time

Answer: B (End-to-end data rate.)

80. In a two-link network, throughput is?

- A) $\text{Max}(R_s, R_c)$
- B) $\text{Min}(R_s, R_c)$
- C) $R_s + R_c$
- D) $R_s * R_c$

Answer: B (Bottleneck link rate.)

81. For N links, throughput without congestion is?

- A) Sum of rates
- B) Min of all rates

C) Average rate

D) Max rate

Answer: B ($\text{Min}(R_1 \text{ to } R_N)$.)

82. In today's Internet, what often constrains throughput?

A) Core links

B) Access network links

C) Global transit

D) No constraint

Answer: B (Access links like R_s and R_c .)

83. In a network with shared core link, what affects throughput?

A) Only access rates

B) Intervening traffic on shared links

C) No effect

D) Propagation only

Answer: B (Shared link divides rate among flows.)

84. If 10 flows share a 5 Mbps link equally, each gets?

A) 5 Mbps

B) 500 kbps

C) 1 Mbps

D) 50 Mbps

Answer: B ($5 \text{ Mbps} / 10 = 500 \text{ kbps}$.)

85. Throughput depends on?

A) Link rates and intervening traffic

B) Delay only

C) Loss only

D) Protocols only

Answer: A (Rates and traffic.)

86. What is the dominant delay in wide-area networks?

A) Processing

B) Propagation

C) Transmission

D) Queuing

Answer: B (Milliseconds over long distances.)

87. What tool gives hands-on feel for end-to-end delay?

A) Ping

B) Traceroute

C) HTTP

D) FTP

Answer: B (Traceroute program.)

88. For a file F over bottleneck $\min(R_s, R_c)$, transfer time is?

- A) $F / \min(R_s, R_c)$
- B) $F * \min(R_s, R_c)$
- C) $\min(F, R_s)$
- D) $R_s + R_c$

Answer: A (*F bits over the rate.*)

89. When is queuing delay zero?

- A) High congestion
- B) Empty queue and no transmission
- C) Full buffer
- D) Traffic intensity > 1

Answer: B (*No wait.*)

90. Packet loss may lead to?

- A) No action
- B) End-to-end retransmission
- C) Faster throughput
- D) Reduced delay

Answer: B (*To ensure data transfer.*)

Section 1.5: Protocol Layers and Their Service Models (Questions 91-110)

91. How are protocols organized?

- A) In a single layer
- B) In layers with service models
- C) Without structure
- D) By hardware only

Answer: B (*Layered architecture.*)

92. What does a layer provide?

- A) Services to the layer above using lower layer services
- B) Hardware only
- C) Protocols only
- D) End systems

Answer: A (*Actions within and using below.*)

93. What are drawbacks of layering?

- A) No duplication
- B) Possible duplication and need for cross-layer info
- C) Always efficient
- D) No separation

Answer: B (*Duplication and violating separation.*)

94. How many layers in the Internet protocol stack?

- A) 7
- B) 5: *physical, link, network, transport, application*
- C) 4
- D) 3

Answer: B (*Five layers.*)

95. What resides in the application layer?

- A) *Routing protocols*
- B) *Protocols like HTTP, SMTP, FTP*
- C) *IP*
- D) *Ethernet*

Answer: B (*Application protocols.*)

96. What is an application-layer packet called?

- A) *Segment*
- B) *Datagram*
- C) *Frame*
- D) *Message*

Answer: D (*Message.*)

97. What does TCP provide?

- A) *Connectionless service*
- B) *Connection-oriented, guaranteed delivery, flow/congestion control*
- C) *No reliability*
- D) *Only routing*

Answer: B (*Reliable service.*)

98. What is a transport-layer packet?

- A) *Datagram*
- B) *Segment*
- C) *Frame*
- D) *Message*

Answer: B (*Segment.*)

99. What does the network layer move?

- A) *Messages*
- B) *Datagrams from host to host*
- C) *Frames*
- D) *Bits*

Answer: B (*Datagrams using IP.*)

100. What protocol is the glue of the Internet?

- A) *TCP*
- B) *IP*
- C) *HTTP*

D) Ethernet

Answer: B (*IP binds the Internet.*)

101. What does the link layer deliver?

A) Datagrams across networks

B) Frames from node to adjacent node

C) Segments

D) Messages

Answer: B (*Over one link.*)

102. Examples of link-layer protocols?

A) IP, TCP

B) Ethernet, WiFi, DOCSIS

C) HTTP, SMTP

D) DNS, FTP

Answer: B (*Link-specific.*)

103. What is a link-layer packet?

A) Segment

B) Datagram

C) Frame

D) Message

Answer: C (*Frame.*)

104. What does the physical layer move?

A) Frames

B) Individual bits within frames

C) Datagrams

D) Segments

Answer: B (*Bits over medium.*)

105. What is encapsulation?

A) Removing headers

B) Adding headers at each layer to the payload from above

C) Switching packets

D) Routing

Answer: B (*Headers added down the stack.*)

106. What layers do routers implement?

A) All five

B) Layers 1-3 (physical, link, network)

C) Only application

D) Transport and above

Answer: B (*Up to network layer.*)

107. What layers do link-layer switches implement?

- A) 1-2 (*physical, link*)
- B) *All*
- C) 3-5
- D) *Only physical*

Answer: A (*Bottom two layers.*)

108. Hosts implement which layers?

- A) *Only bottom*
- B) *All five*
- C) *Network only*
- D) *Link and physical*

Answer: B (*All layers.*)

109. In encapsulation, what is the payload?

- A) *Header only*
- B) *Packet from the layer above*
- C) *Bits*
- D) *No data*

Answer: B (*Upper layer packet.*)

110. What is UDP?

- A) *Connection-oriented*
- B) *Connectionless with no reliability or control*
- C) *Routing protocol*
- D) *Application protocol*

Answer: B (*No-frills service.*)

Section 1.6: Networks Under Attack (Questions 111-120)

111. What is malware?

- A) *Beneficial software*
- B) *Malicious software that infects devices*
- C) *Network hardware*
- D) *Protocols*

Answer: B (*Does harm like deleting files or spying.*)

112. What is a botnet?

- A) *Single compromised device*
- B) *Network of compromised devices for attacks*
- C) *Secure network*
- D) *Protocol stack*

Answer: B (*Used for spam or DDoS.*)

113. How does malware spread?

- A) Non-self-replicating*
- B) Self-replicating from infected hosts*
- C) Only via hardware*
- D) No spread*

Answer: B (*Exponentially fast.*)

114. What is a DoS attack?

- A) Enhances service*
- B) Renders network/host unusable*
- C) Improves throughput*
- D) Reduces delay*

Answer: B (*Denial-of-service.*)

115. Types of DoS attacks include?

- A) Vulnerability, bandwidth flooding, connection flooding*
- B) Only packet loss*
- C) Queuing only*
- D) Encryption*

Answer: A (*Three categories.*)

116. What is a DDoS attack?

- A) Single source*
- B) Distributed from multiple sources*
- C) No flooding*
- D) Easy to detect*

Answer: B (*Harder to defend.*)

117. What is packet sniffing?

- A) Injecting packets*
- B) Passively recording packets*
- C) Encrypting data*
- D) Routing*

Answer: B (*Copies packets for analysis.*)

118. Why are wireless channels vulnerable to sniffing?

- A) Wired only*
- B) Broadcast nature*
- C) No packets*
- D) Encryption always*

Answer: B (*Anyone in range can capture.*)

119. What is IP spoofing?

- A) True source address*
- B) Injecting packets with false source address*
- C) Secure authentication*

D) No masquerading

Answer: B (*Masquerading as another.*)

120. What solves IP spoofing?

A) No mechanism

B) End-point authentication

C) More delays

D) Packet loss

Answer: B (*Verifies origin.*)

Section 1.7: History of Computer Networking and the Internet (Questions 121-135)

121. When was the ARPAnet operational?

A) 1960s

B) Late 1960s (1969)

C) 1980s

D) 1990s

Answer: B (*First packet-switched network.*)

122. What was the first message on ARPAnet?

A) Hello world

B) LO (attempted LOGIN)

C) Packet sent

D) Internet

Answer: B (*Crashed after LO.*)

123. What protocols were in place by the 1970s end?

A) HTTP, SMTP

B) TCP, UDP, IP

C) Ethernet, WiFi

D) DNS, FTP

Answer: B (*Key Internet protocols.*)

124. How many hosts on ARPAnet by 1970s end?

A) Millions

B) About 200

C) Thousands

D) Billions

Answer: B (*Approximately 200.*)

125. When was TCP/IP officially deployed?

A) 1970

B) January 1, 1983

C) 1990

D) 2000

Answer: B (*Replaced NCP.*)

126. What was the Minitel project?

A) US effort

B) French data networking to homes in 1980s

C) Satellite network

D) Mobile access

Answer: B (*In French homes before Internet awareness.*)

127. How many hosts on Internet by 1980s end?

A) 200

B) 100,000

C) 1 million

D) 1 billion

Answer: B (*Reached 100,000.*)

128. When was the DNS developed?

A) 1970s

B) 1980s (RFC 1034)

C) 1990s

D) 2000s

Answer: B (*For name to IP mapping.*)

129. What started the browser war?

A) Google

B) Microsoft making browsers in 1996

C) Netscape only

D) Amazon

Answer: B (*Microsoft vs. Netscape.*)

130. Killer applications by millennium end?

A) Only email

B) Email, Web, instant messaging, P2P file sharing

C) Streaming only

D) Social networks

Answer: B (*Four key apps.*)

131. Big winners in Internet space?

A) Only startups

B) Microsoft, Cisco, Yahoo, eBay, Google, Amazon

C) Telcos only

D) Governments

Answer: B (*Major companies.*)

132. What enabled user-generated video?

- A) Low-speed access
- B) Broadband like cable/DSL/FTTH/5G
- C) No wireless
- D) Circuit switching

Answer: B (*High-speed access.*)

133. When did wireless devices surpass wired on Internet?

- A) 2000
- B) 2011
- C) 2020
- D) Never

Answer: B (*In 2011.*)

134. What are online social networks examples?

- A) Email only
- B) Facebook, Instagram, Twitter, WeChat
- C) Routers
- D) Protocols

Answer: B (*For messaging and sharing.*)

135. Where do many apps run now?

- A) Local devices only
- B) In the cloud like Amazon EC2, Azure
- C) No networks
- D) Circuit switches

Answer: B (*Cloud providers.*)

Section 1.8: Summary (Questions 136-140)

136. What is the roadmap sequence?

- A) Security first
- B) Chapters on layers: application, transport, network (data/control), link, then wireless/security
- C) History only
- D) Physical first

Answer: B (*Chapters 2-6 on top layers, 7-8 on wireless/security.*)

137. What chapters are the core?

- A) 1 and 8
- B) 2 through 6
- C) 7 and 8
- D) All

Answer: B (*Organized around top four layers.*)

138. What does Chapter 7 cover?

- A) Application layer*
- B) Wireless and mobile networks (WiFi, cellular, mobility)*
- C) Security*
- D) History*

Answer: B (*Wireless topics.*)

139. What does Chapter 8 address?

- A) Delays*
- B) Security in networks, encryption, applications*
- C) Throughput*
- D) Protocols*

Answer: B (*Security underpinnings and applications.*)

140. Where is complexity in Internet architecture?

- A) Core only*
- B) At the edges (end systems)*
- C) Links*
- D) No complexity*

Answer: B (*Puts complexity at edges.*)

Wish you all the best!

Fathy.